

JONATHAN HUANG

647-896-8607 | j664huan@uwaterloo.ca | Toronto, ON | jhuang0701.github.io/portfolio

SKILLS

- Languages & Frameworks: Python, C++, JavaScript, HTML, CSS, React, Tailwind CSS, Streamlit, Assembly
- Tools & Platforms: Git, JIRA, PostgreSQL, Supabase, REST APIs, Power Automate, Linux/Bash, MATLAB, Verilog
- Data & AI: Pandas, Plotly, pgvector, statistical modeling, LLM API integration, Groq API, RAG pipelines
- Embedded & Hardware: Arduino, ESP32, AM32 ESC programming, embedded C++, I2C/SPI, AutoCAD, SolidWorks

EDUCATION

University of Waterloo, Waterloo, ON

Sep 2025 – Aug 2030

Bachelor of Applied Science in Electrical Engineering, 91% average, 4.0 GPA

- Awards: University of Waterloo President's Scholarship of Distinction (2025)

EXPERIENCE

S&C Electric Company, Toronto, ON

Assistant QA Engineer

Jan – Apr 2026

- Reduced document retrieval time by 60% for 1,000+ technical records by engineering automated Python data pipelines with custom validation scripts and CI/CD-style workflows to streamline database migration
- Improved inspection traceability and reduced audit discrepancies by 40% through development of Python and HTML certification logging apps with unit-tested data validation modules and pytest-style process automation
- Eliminated manual calibration bottlenecks across 500+ precision tools by designing intelligent Power Automate and GAGEpack workflows, achieving real-time compliance tracking across measurement systems
- Ensured zero non-conformance escapes across 500+ OMNI Switchgear and Vista Motor systems by executing structured QA test plans against engineering blueprints and maintaining traceable defect logs

University of Waterloo WATBOTs, Waterloo, ON

Electrical Subteam Lead

Sep 2025 – Present

- Delivered a fully functioning competition combat robot within two academic terms by leading a cross-functional team of 20+ engineers across system architecture, power distribution, motor control, and failsafe integration
- Delivered a competition-ready electrical system within a 3lb weight limit, 85°C thermal threshold, and 5V/20mA ESP32 constraints by designing optimized circuit schematics validated through load testing and current analysis
- Enabled precise real-time motor control by developing embedded C++ firmware on Arduino, programming AM32 ESCs for brushless motor operation, and integrating an ESP32-based receiver for wireless command processing
- Performed hands-on electrical assembly including precision soldering and wire harnessing, collaborating with mechanical and software teams to ensure full system integration and competition compliance

Act First Safety, Toronto, ON

Administrative Intern

Jul – Sep 2024

- Improved data accuracy and retrieval speed for 2,000+ customer contracts by building Excel macro-based validation pipelines, reducing lookup errors and enabling company-wide accessibility
- Resolved a multi-month administrative backlog within 8 weeks by optimizing data entry workflows, expense reconciliation, and document compliance filing processes

PROJECTS

ChatTFT, Teamfight Tactics (TFT) AI Analyzer

Jan 2026 – Present

Python, PostgreSQL, Streamlit, Supabase, Pgvector, RAG, LLM evaluation | <https://chattft.streamlit.app>

- Built and deployed a Python-based AI analytics platform integrating Llama 3.3 70B via Groq with a production RAG pipeline using Supabase pgvector, semantic chunking, and IVFFlat-indexed retrieval over 10K+ chunks
- Designed a statistical data pipeline processing thousands of matches using Laplace-smoothed models to benchmark player performance and identify meta gaps, serving 300+ users and improving win rates by 40%+
- Implemented a context-aware AI chat system with multi-turn memory and intelligent query routing, augmented by an LLM-as-judge eval framework and prompt A/B testing across 3 variants, improving actionability scores by 18%
- Deployed a full-stack production application backed by a PostgreSQL database, multi-turn memory, and a fine-tuning dataset generator producing 2K+ labeled examples, receiving official Riot Games API product approval

Warm Wheels, Hot Wheels Launcher Toy

Feb – Jun 2025

- Designed and deployed an embedded control system using AutoCAD for mechanical design, integrating a pulley-driven actuator with brushed DC motors, an L298N motor driver, and an LCD interface for real-time user feedback
- Developed embedded C++ firmware on Arduino Uno Rev3 implementing a real-time sequencing algorithm with state machine logic for motor control synchronization, input validation, and hardware-software integration